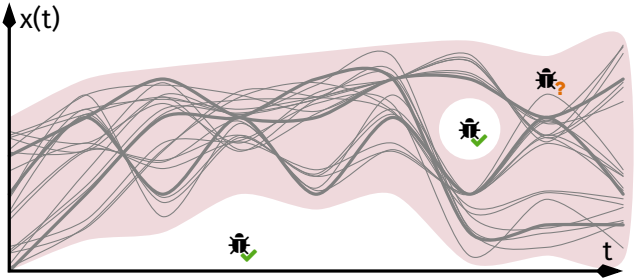


Abstract Interpretation

Software Quality Assurance — Static Code Analysis, II | Florian Sihler | December 10, 2025

Software Engineering
Programming Languages

universität ulm

Outline

1. The Why

2. The How

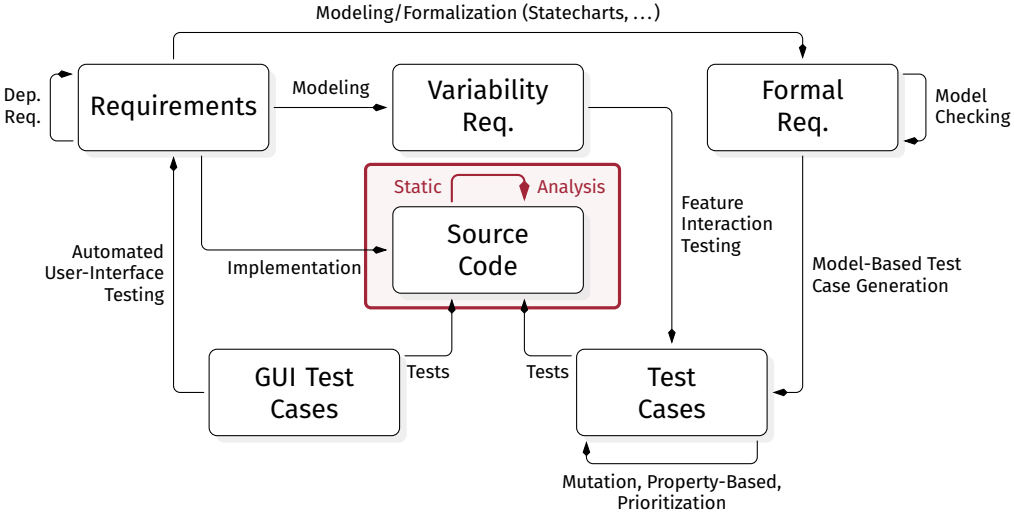
3. Semantics

4. Outlook and Comments

1. The Why

Why am I even here?

Embedding a Landscape



What is static analysis?

Discover *syntactic/semantic properties* of programs **without** running them.[RY20]

Today, we learn how to…

- describe semantic properties
- compare, refine, and combine properties
- describe and map language semantics to properties
- deal with the cost of abstraction (and the fun)

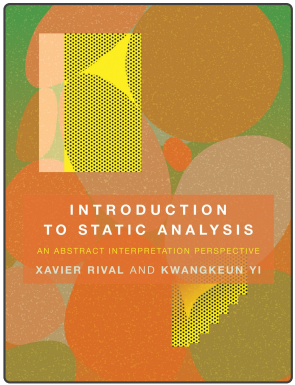
The Why

```java
public static void main(String[] args) {
   int a = 1;                          ⦇ a ∈ {1} ⦈
   double r = Math.random() * 10;      ⦇ r ∈ [0 .. 10) ⦈
   if (r > 5) {
      a = 2;                           ⦇ a ∈ {2} ⦈
   }
   System.out.println(1 / a);          ⦇ a ∈ {1, 2} ⦈ → Valid? Ok? Safe?
}
```

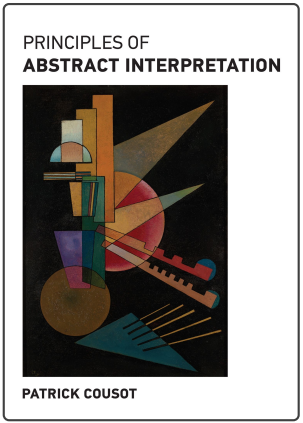
- We want to proof, that a program satisfies certain properties
- Abstract Interpretation is one (/the) technique to do so

Recommended Resources

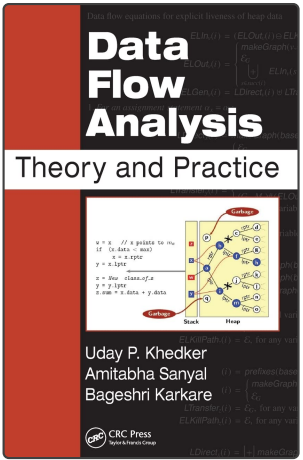
Using Analyses [RY20]



Formal Foundations [Cou21]



Dataflow Perspective [KSK09]



And for an overview: "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17]

Recommended Resources

Using Analyses [RY20] Formal Foundations [Cou21] Dataflow Perspective [KSK09]

And for an overview: "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17]

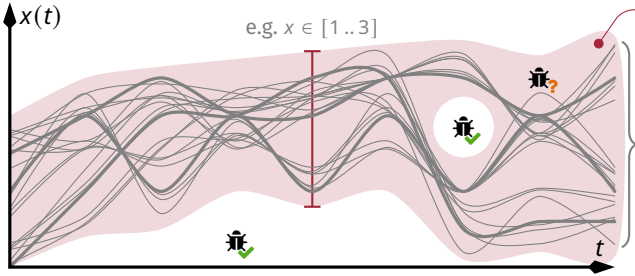
2. The How

Gimme properties, gimme abstractions!

Abstract Interpretation

Radhia Cousot (1947–2014)
Patrick Cousot

Patrick Cousot (1948)
Personal Website



(Trace) Abstraction[Cou21, p. 92]
just one of many
this **must** include all concrete traces!
(over-approximate)

Collecting Semantics[Cou21, p. 91]

- Maybe impossible to compute statically
- Each trace is a single execution (test, …)
- … or very expensive (▸ *dynamic*)
- ▸ Abstract Interpretation to the rescue

See "A casual introduction to Abstract Interpretation" [Cou12]

Terminology

- **Property** — Set of states/traces that satisfy that property
  Even integers: $\mathsf{P} = \{\, z \in \mathbb{Z} \mid \exists k \in \mathbb{Z} : z = 2k \,\} = \{\, 0, 2, -2, 4, -4, 6, \ldots \,\} \subseteq \mathcal{P}(\mathbb{Z})$ universe ($\mathbb{U}$)

$$\emptyset \subseteq \mathsf{P}_1 \subseteq \mathsf{P}_2 \subseteq \mathbb{U}$$

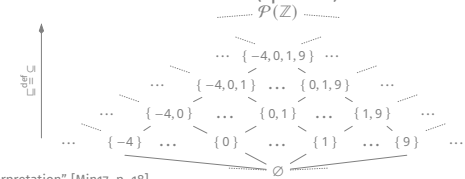
strongest, stronger, weaker, weakest

$$\forall x, y, z \in X : x \sqsubseteq y \wedge y \sqsubseteq z \implies x \sqsubseteq z$$

$$\forall x \in X : x \sqsubseteq x \qquad \forall x, y \in X : x \sqsubseteq y \wedge y \sqsubseteq x \implies x = y$$

- **Partial Order** — A reflexive, transitive, antisymmetric relation on a set ("poset")
  $(\mathbb{Z}, \leq), \quad (\mathcal{P}(\mathbb{Z}), \subseteq), \quad \ldots$

Hasse Diagram of $(\mathcal{P}(\mathbb{Z}), \subseteq)$



"Principles of Abstract Interpretation" [Cou21, p. 15],"Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 18]

A Task In-Between

Helmut Hasse (1898–1979)
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Photo Collection

Task:
Consider the Poset $(\mathcal{P}(\{0, 3, -2\}), \subseteq)$.
Draw its Hasse diagram.
Indicate the greatest and smallest element.

Hasse Diagram:
A directed graph, with edges from a to b ($a \neq b$) indicating that $a \sqsubseteq b$ without a c with $a \sqsubseteq c \sqsubseteq b$.

A Task In-Between

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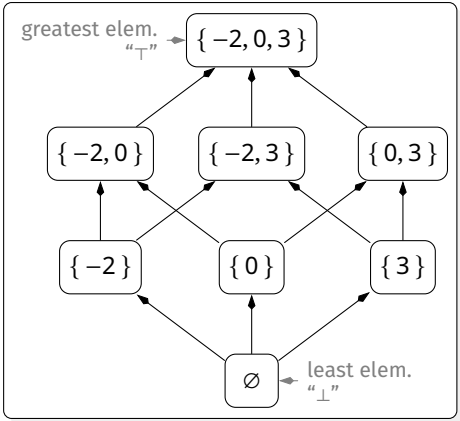
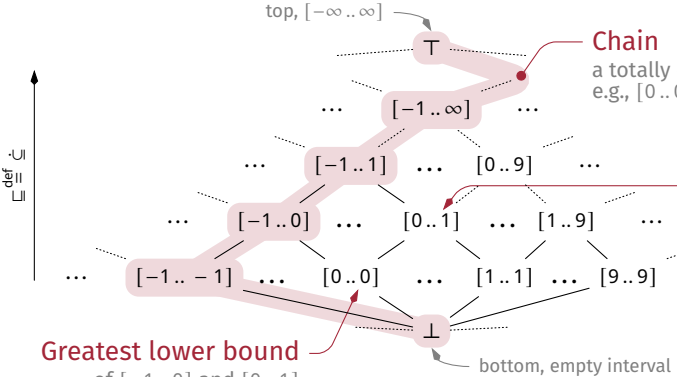


Hasse Diagram of $(\mathcal{P}(\{\mathbf{0}, \mathbf{3}, -\mathbf{2}\}), \subseteq)$

Posets, Lattices, and Chains

Garrett Birkhoff (1911–96)
Paul Halmos



Chain
a totally ordered subset
e.g., $[0\,..\,0] \sqsubseteq [0\,..\,9] \sqsubseteq [-10\,..\,200]$

Least upper bound
of $[0\,..\,0]$ and $[1\,..\,1]$
lub, join, $\sqcup$

Greatest lower bound
of $[-1\,..\,0]$ and $[0\,..\,1]$
glb, meet, $\sqcap$

"Lattice elements (e.g. $[0\,..\,1]$) define — per variable — the abstract state (the abstraction) at a given time or program point!"

"Lattice theory" [Bir67], see also sublattices [Min17, p. 25]

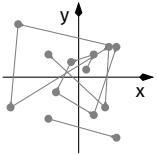
Complete Lattice $(X, \sqsubseteq, \sqcup, \sqcap, \bot, \top)$

- $(X, \sqsubseteq)$ is a partial order
- $\forall A \subseteq X : \sqcup A$ and $\sqcap A$ exist
- $\bot/\top$ as smallest/largest element

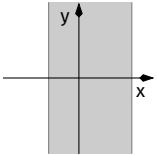
Abstract Domains

Pick Your Favorite Lattice!

Numerical

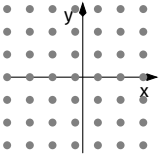


Collecting Semantics

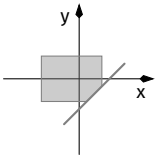


Intervals $x \in [a \,..\, b]$

$y \in [-\infty \,..\, \infty]$



Simple Congruences



Pentagons

- Octagons
- Ellipses
- Exponentials
- Signs

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

Sign Analysis

Simple Sign Domain

- We still have no program semantics, but we can try…

```
int a = 0;
int b = 12;
int c = a + b;
int d = c - b;
```

java

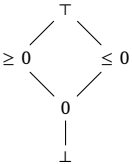
$\wr a = 0 \int$

$\wr b \geq 0 \int$

$\wr c \geq 0 \quad (= 0 \ + \ \geq 0) \int$

$\wr d = \top \quad (\geq 0 \ - \ \geq 0) \int$

- But, how do we know that `int a = 0;` implies $\wr a = 0 \int$?
  - Language Semantics (we get to those later)
  - Galois Connections

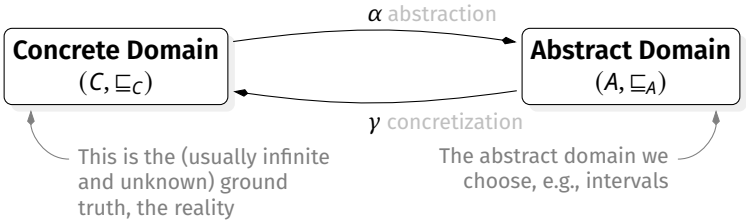


Galois Connection — Linking Posets [Cou21, p. 110] (small detour)

Évariste Galois (1811–32)
Public Domain
died in a duel for love

How do we know that `int a = 0;` implies $\wr\, a = 0 \,\wr$?

A tale of two lattices...



Beware, beware, conversions may be lossy!

- $\forall x \forall y : \alpha(x) \sqsubseteq_A y \iff x \sqsubseteq_C \gamma(y)$
- $\forall x \forall y : x \sqsubseteq_C \gamma(\alpha(x))$ and $\alpha(\gamma(y)) \sqsubseteq_A y$

"We can abstract $x \in \{1, 2\}$ with $\alpha(x) = [1 \,..\, 2]$ or $\alpha(x) = [-5 \,..\, 42]$, but *not* (e.g.) with $\alpha(x) = [1 \,..\, 1]$! "

Happy Abstractions

- Suppose, you use the interval domain, what are your best abstractions if u is an unknown integer?
  - $\alpha(\texttt{u}) =$
  - $\alpha(\texttt{u + 1}) =$
  - $\alpha(\texttt{u * 0}) =$
  - $\alpha(\texttt{sin(u)}) =$
- What is the best concretization of $[-5\,..\,5]$? What do we loose?

Happy Abstractions

- Suppose, you use the interval domain, what are your best abstractions if u is an unknown integer?
  - $\alpha(\texttt{u}) = [-\infty \,..\, \infty]$
  - $\alpha(\texttt{u + 1}) = [-\infty \,..\, \infty]$
  - $\alpha(\texttt{u * 0}) = [0 \,..\, 0]$
  - $\alpha(\texttt{sin(u)}) = [-1 \,..\, 1]$
- What is the best concretization of $[-5 \,..\, 5]$? What do we loose?
  $\gamma([-5 \,..\, 5]) = \{-5, -4, \ldots, 4, 5\}$. We loose...
  - Exact values and distribution information
  - Relational information (e.g., `x = u; y = x + 1;`)
  - Potential sequences (e.g., `for(i=0; i<n; i++) ...`)
- But, how do we handle loops? Recursion?

Fixpoints

Alfred Tarski (1901–83)
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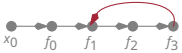
- For operators $f : X \to X$ a **fixpoint** is a $x \in X$ such that $f(x) = x$
- If we iterate f starting from some $x_0 \in X$:
  -

    reach a fixpoint, $f^p = f(f^p)$
  - 

    reach a cycle, $f^{p+\ell} = f^p, \ell > 0$
  -

    iterate forever, $\forall p \neq q \in \mathbb{N} : f^p \neq f^q$

$f : \mathbb{N} \to \mathbb{N}, f(x) = x + 1$

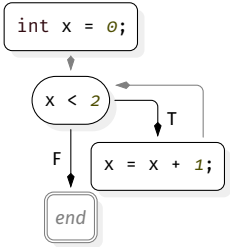
- If our function is monotonic, we can always find a fixpoint[Tar55]

for complete, nonempty lattices
Tarski's Theorem

- Analyzing, e.g. loops, we "go up" the lattice until we reach a least fixpoint
- You may know fixpoints from tools like LaTeX or the λ-calculus

"A lattice-theoretical fixpoint theorem and its applications." [Tar55], "Introduction to metamathematics" [Kle52], "Principles of Abstract Interpretation" [Cou21, p. 165]

Interval Analysis, I (the intuitive approach)



$\wr\, x_0 \in [0\,..\,0] \,\wr$

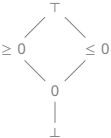
$\wr$ [pre] $x_1 \in [0\,..\,0] \,\wr$

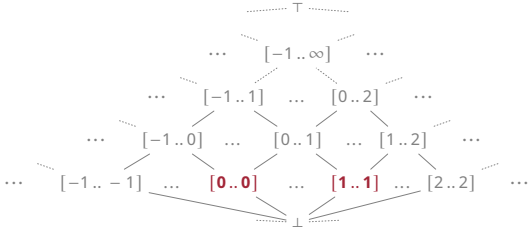
$\wr$ [in] $x_2 \in [0\,..\,0] \quad ([0\,..\,0] \cap (-\infty\,..\,1]) \,\wr$

if we are inside the loop we know that $x < 2$ holds!

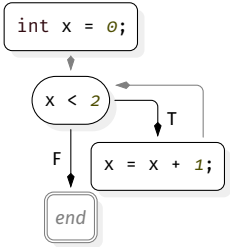
$\wr\, x_3 \in [1\,..\,1] \quad ([0\,..\,0] \oplus [1\,..\,1]) \,\wr$

we have to know how to add intervals here





Interval Analysis, I (the intuitive approach)



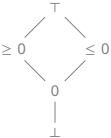
$\wr\, x_0 \in [0\,..\,0] \,\wr$

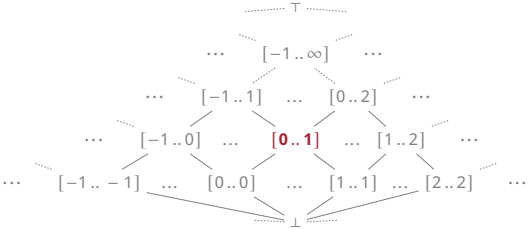
we join on loops because, at this point x can have any of the values

$\wr$ [pre] $\boldsymbol{x_1 \in [0\,..\,1]} \quad \boldsymbol{([0\,..\,0] \cup [1\,..\,1])} \,\wr$

$\wr$ [in] $x_2 \in [0\,..\,1] \quad ([0\,..\,1] \cap (-\infty\,..\,1]) \,\wr$

$\wr\, x_3 \in [1\,..\,2] \quad ([0\,..\,1] \oplus [1\,..\,1]) \,\wr$





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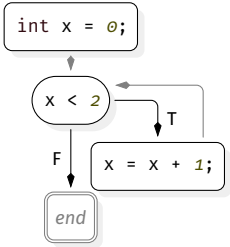


Intervals

$\wr\, x_0 \in [0\,..\,0] \,\wr$

$\wr$ [pre] $\boldsymbol{x_1 \in [0\,..\,2]}$ $\quad$ $\boldsymbol{([0\,..\,1] \cup [1\,..\,2])} \,\wr$

$\wr$ [in] $x_2 \in [0\,..\,1]$ $\quad$ $([0\,..\,1] \cap (-\infty\,..\,1]) \,\wr$

$\wr\, x_3 \in [1\,..\,2]$ $\quad$ $([0\,..\,1] \oplus [1\,..\,1]) \,\wr$

$\wr$ [post] $x_4 \in [2\,..\,2]$ $\quad$ $([0\,..\,2] \cap [2\,..\,\infty)) \,\wr$

after the loop we know that $\neg(x < 2) = x \geq 2$ holds!

Signs

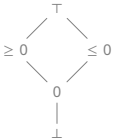
$\wr\, x_0 = 0 \,\wr$

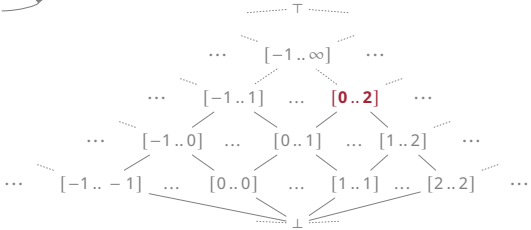
$\wr$ [pre] $x_1 \geq 0 \,\wr$

$\wr$ [in] $x_2 \geq 0 \,\wr$

$\wr\, x_3 \geq 0 \,\wr$

$\wr$ [post] $x_4 \geq 0 \,\wr$





3. Semantics

What does my program mean?

Program Syntax (simplified)

```
int x = 0;
while(x < 2) {
  x = x + 1;
}
```

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

Comparison $\bowtie \in \{\leq, <, \ldots\}$

Binary Expression

stm	::=	$V \leftarrow$ *expr*	(assignment, $V \in \mathbb{V}$)	java
	|	stm_1; stm_2	(sequence)	
	|	**while**(*cond*) { *stm* }	(loop)	
expr	::=	V	(variable, $V \in \mathbb{V}$)	
	|	c	(constant, $c \in \mathbb{I}$)	
	|	$expr_1 \diamond expr_2$	(bin. expr., $\diamond \in \{+, -, \ldots\}$)	
cond	::=	b	(boolean, $b \in \mathbb{B}$)	
	|	$expr_1 \bowtie expr_2$	(comparison, $\bowtie \in \{\leq, <, \ldots\}$)	

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

$$
\begin{array}{lcll}
\mathit{expr} & ::= & V & (\text{variable}, V \in \mathbb{V}) \\
 & | & c & (\text{constant}, c \in \mathbb{I}) \\
 & | & \mathit{expr}_1 \diamond \mathit{expr}_2 & (\text{bin. expr.}, \diamond \in \{+, -, \ldots\})
\end{array}
$$

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

$\mathbb{V}$	$\mathbb{I}$
x	0
c	5

Usually written as $\mathbb{E}[\![\, \mathit{expr} \,]\!]\rho$

- Now we can define evalExpr(expr, env) for an environment $\mathit{env} \in \mathcal{E}$

$$
\begin{aligned}
\text{evalExpr}(V,\ \mathit{env}) &\stackrel{\text{def}}{=} \mathit{env}(V) \\
\text{evalExpr}(c,\ \mathit{env}) &\stackrel{\text{def}}{=} c \\
\text{evalExpr}(\mathit{expr}_1 + \mathit{expr}_2,\ \mathit{env}) &\stackrel{\text{def}}{=} \text{evalExpr}(\mathit{expr}_1,\ \mathit{env}) + \text{evalExpr}(\mathit{expr}_2,\ \mathit{env}) \\
&\ \vdots
\end{aligned}
$$

Value of $V \in \mathbb{V}$ in Environment env

$\mathbb{C}[\![\, \mathit{cond} \,]\!]\mathcal{D}$

$\mathbb{S}[\![\, \mathit{stm} \,]\!]\mathcal{D}$

- Additionally, we can define evalCond(cond, envs) and evalStm(stm, envs)

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

Denotational Semantics while loops

Suppose we start the loop with states *Start*

$$\textbf{while}(cond)\ \{\ stm\ \} \qquad F(X) \stackrel{\text{def}}{=} Start \cup evalStm(stm,\ evalCond(cond,\ X))$$

iterate to find the least fixpoint [Min17, p. 52]

Try to map this to the *Interval Analysis, I* example!

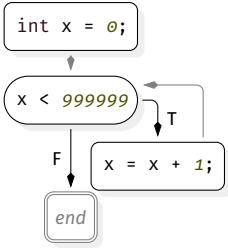
Keep only states S with $evalCond(\neg cond,\ S)$

There are alternatives (e.g., equation systems, [Cou21, part 7])

We achieve their abstract counterpart using the same principles but for abstract domains!

Usually written as $\mathbb{S}^\#, \mathbb{C}^\#, \mathbb{E}^\#, \ldots$

Interval Analysis, II



$\wr\, x_0 \in [0\,..\,0] \,\wr$

$\wr$ [pre] $x_1 \in [0\,..\,2] \quad ([0\,..\,1] \cup [1\,..\,2]) \,\wr \;\; \nabla \implies x_1 \in [0\,..\,\infty)$

$\wr$ [in] $x_2 \in [0\,..\,1] \quad ([0\,..\,1] \cap (-\infty\,..\,1]) \,\wr$

$\wr\, x_3 \in [1\,..\,2] \quad ([0\,..\,1] \oplus [1\,..\,1]) \,\wr$

$\wr$ [post] $x_4 \in [999999\,..\,\infty) \quad ([0\,..\,\infty) \cap [999999\,..\,\infty)) \,\wr$

- Fixpoint iteration can be *very* expensive, and may not stabilize
- *Widening* (∇) is crucial, computing an upper bound
  (We only need widening if the lattice has an infinite ascending chain!)

"Principles of Abstract Interpretation" [Cou21, p. 393], "Widening and narrowing operators for abstract interpretation" [CZ11]

Let's Bring it All Together

Sign Analysis

I want to make a sign analysis! (amazing!)

1. Define the lattice! (there are many solutions)
   drawing the hasse diagram is enough for us here

2. Define the abstract semantics on the following language!

$$
\begin{array}{lcll}
\mathit{expr} & ::= & c & \text{(constant, } c \in \mathbb{R}\text{)} \\
 & | & \mathit{expr}_1 + \mathit{expr}_2 & \text{(addition)}
\end{array}
$$

3. Provide an integer concretization for $\gamma(0)$ and $\gamma(\leq 0)$!

4. Do we need widening? If so, define ∇!

Let's Bring it All Together — Sign Analysis

I want to make a sign analysis! (amazing!)

$$\begin{tikzcd} & \geq 0 \arrow[dr, dash] & \\ \top \arrow[ur, dash] \arrow[dr, dash] & & 0 \arrow[r, dash] & \bot \\ & \leq 0 \arrow[ur, dash] & \end{tikzcd}$$

1. Define the lattice! (there are many solutions)
   drawing the hasse diagram is enough for us here
   $\mathcal{L} = (x = \{\bot, 0, \leq 0, \geq 0, \top\},\ \leq = \{(\bot, 0), (\bot, \leq 0), \ldots\}, \sqcup = \{(\bot, x) \mapsto x, (0, \leq 0) \mapsto \leq 0, \ldots\},$
   $\sqcap = \{(\top, x) \mapsto x, (0, \leq 0) \mapsto 0, \ldots\}, \bot, \top)$

2. Define the abstract semantics on the following language!

   $$\begin{array}{lcll} \mathit{expr} & ::= & c & \text{(constant, } c \in \mathbb{R}\text{)} \\ & \mid & \mathit{expr}_1 + \mathit{expr}_2 & \text{(addition)} \end{array}$$

   $$\mathbb{E}[\![\, c \,]\!]\rho \stackrel{\text{def}}{=} \begin{cases} \geq 0 & \text{if } c > 0 \\ 0 & \text{if } c = 0 \\ \leq 0 & \text{if } c < 0 \end{cases}$$

   short for evalExpr($\mathit{expr}, \rho = \mathit{env}$)

   $$\mathbb{E}[\![\, a + b \,]\!]\rho \stackrel{\text{def}}{=} \mathbb{E}[\![\, a \,]\!]\rho \oplus \mathbb{E}[\![\, b \,]\!]\rho$$

$\oplus$	$\bot$	0	$\leq$	$\geq$	$\top$
$\bot$	$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
0		0	$\leq$	$\geq$	$\top$
$\leq$			$\leq$	$\top$	$\top$
$\geq$				$\geq$	$\top$
$\top$					$\top$

   $x \oplus y = y \oplus x$

3. Provide an integer concretization for $\gamma(0)$ and $\gamma(\leq 0)$!
   $\gamma(0) = \{0\}, \quad \gamma(\leq 0) = \{n \in \mathbb{Z} \mid n \leq 0\}$

4. Do we need widening? If so, define ∇!
   We do not need widening here as our lattice is finite and our semantics do not introduce new elements. (We will concretize this next week.)

4. Outlook and Comments

This is incredible, I need more!

The Domains We Built…

- We know a handful of important concepts for our domains:
  - Lattices (poset, with join $\sqcup$, meet $\sqcap$, bottom $\bot$, and top $\top$)
  - Program Semantics ($\mathbb{S}[\![\mathit{stm}]\!]\mathcal{D}$, …, e.g., $\mathbb{E}[\![a + b]\!]\rho = \mathbb{E}[\![a]\!]\rho \oplus \mathbb{E}[\![b]\!]\rho$)
  - Fixpoint Iterations (to interpret loops, recursion, …)
  - Widening (to ensure termination with infinite chains)
  - Galois Connections (to relate concrete and abstract domains)
- With abstract interpretation we interpret programs *over* these domains!
- However, we have only looked at single variables and single domains so far!

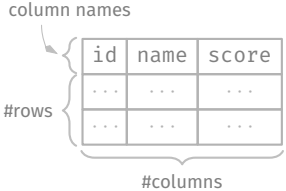
Combining Domains: Products

Saunders Mac Lane (1909–2005)
🅭🅯🄏 2.0 Oberwolfach Photo Collection

Samuel Eilenberg (1913–1998)
🅭🅯🄏 2.0 Oberwolfach Photo Collection

- Let's assume we have two domains D_1 and D_2
- We can combine them into a *product domain*: $D_1 \times D_2$

- For example, let's suppose we want to track the shape of tables! We define three domains!
  - $\mathcal{N} \stackrel{\text{def}}{=} \mathcal{P}(\Sigma^*)$ (set of column names)
  - $\mathcal{R} = \mathcal{C} \stackrel{\text{def}}{=} \{\, [a\,..\,b] \mid a, b \in \mathbb{N}_0 \cup \{\infty\}, a \leq b \,\}$ (# of rows/columns)
  - Then we can define the domain: $\mathcal{DF} \stackrel{\text{def}}{=} \mathcal{N} \times \mathcal{R} \times \mathcal{C}$



[Ger25] Oliver Gerstl. Tracking the shape of data frames in R programs using abstract interpretation (Ulm University, 2025)

Category theory is amazing!

Outlook

- Domain transformers
  combine abstract domains [Min17, p. 149]
- Galois connections (offer so much more)
  define the relationship between concrete and abstract domains [Cou21, p. 110]
- Corresponding to widening, narrowing
  refines approximations [Cou21, p. 395]
- Function calls
  require special handling [MJ12]
- Existing libraries allow for easy implementation
  LiSA [Fer+21], MOPSA [Jou+19], Apron [JM09]
- There are other ways to define semantics
  e.g., small-step, big-step [Cio13]

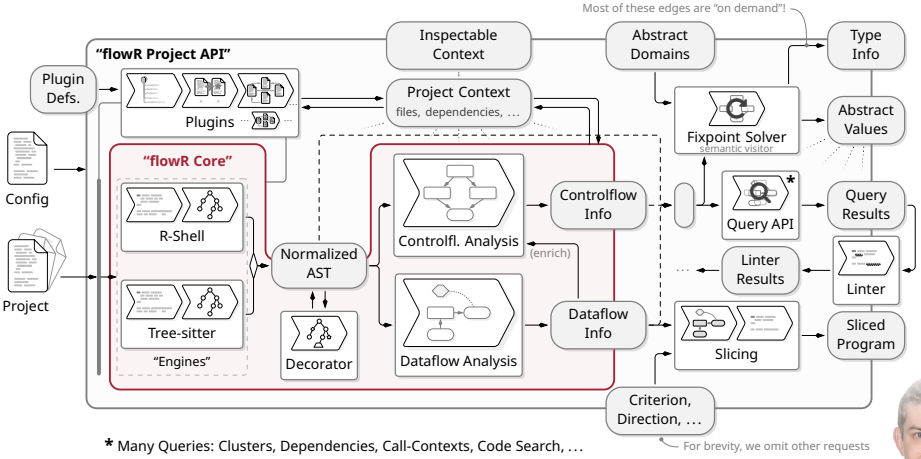
Domains can also capture relations between variables (e.g. *polyhedra*), their provenance, and much more!

However, this implies trade-offs which we discuss next time.

Back to the Questions

1. How would you capture what a *property* is? ✔
2. How would you phrase that one property is “better” than another? ✔
3. For what operations would you *not* use a control-flow graph?
4. Why can’t there be a fully automatic, sound, and complete static analyzer for general programs?
5. What (big) additional challenges do you see in the real-world?

And… Back to the Real World?



* Many Queries: Clusters, Dependencies, Call-Contexts, Code Search, …

In case you like/are intrigued by what you see, join the horde: **florian.sihler@uni-ulm.de**

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A Little Bit of History

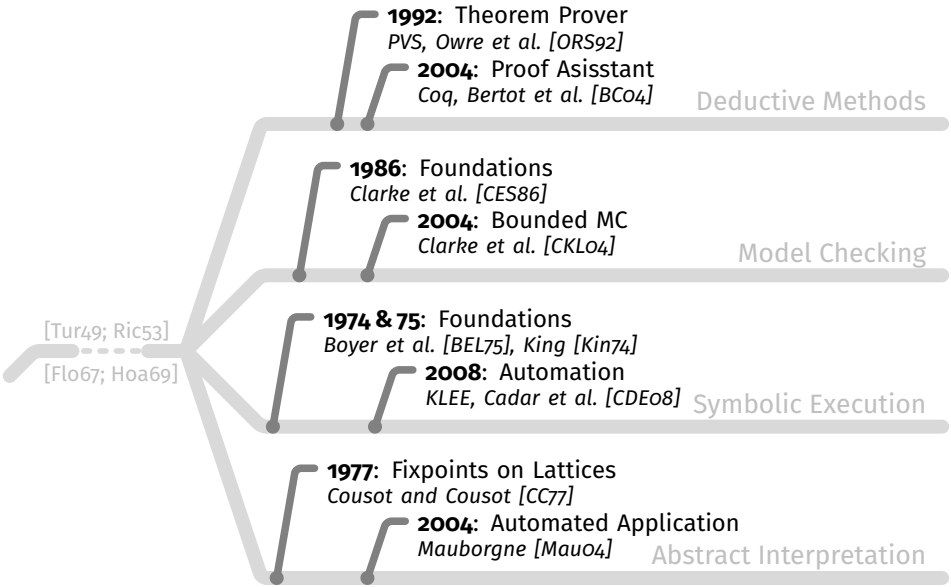
Static Analysis

1949: First Checks
Turing [Tur49]

1953: Rice Theorem
Non-trivial properties are undecidable [Ric53]

1967 & 69: Logical Foundation
Floyd [Flo67], Hoare [Hoa69]
But: No Automation

Based on the amazing "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" by Miné [Min17], https://www.di.ens.fr/~cousot/AI/, and [Bal+18; GR22]



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